

July 16, 2026 03:58 PM GMT

TSMC | Asia Pacific

Aggressive capex to address stronger AI demand; OW

WHAT'S CHANGED

TSMC (2330.TW)	From	To
Price Target	NT\$2,888.00	NT\$2,988.00

Stronger capex points to larger and more urgent AI infra demand. TSMC remains the primary, if not the only, 2nm/1.4nm foundry option for major customers over the next five years.

2026 full-year revenue guidance a strong beat: TSMC hosted its 2Q26 earnings call today. It raised its 2026 revenue growth guidance to >40% Y/Y (previously >30% Y/Y). In our 6/28 [preview](#), we expected guidance to increase to 40% Y/Y vs. Consensus 35%-40%. Management attribute upside to strong AI demand despite challenging consumer demand. CSP customers are rapidly increasing cloud capex. TSMC did not update its AI semiconductor revenue CAGR forecast but indicated it is tracking above its previous 55%–60% forecast. We believe a 70%–80% CAGR for TSMC's AI semiconductor business is reasonable.

Capex hike implies urgent AI demand: TSMC raised its 2026 capex target to US \$60–64bn (vs. the previous US\$52–56bn), implying some equipment cost inflation. We attribute the additional US\$8bn to: (1) US\$3bn of cost inflation (reflecting a 5% increase in equipment prices), and (2) US\$5bn of equipment prepayments, given that TSMC is not adding new clean-room capacity this year. These equipment prepayments suggest urgent AI demand, particularly for agentic AI-related CPUs and GPUs. Management said A14 would be an even larger node than 2nm and announced an additional US\$100bn of capex for its US fabs, which we believe supports higher revenue after 2028 than we previously expected.

Do we need to worry about gross margin "miss" and foundry competition? In response to our question on competition from Samsung Foundry and Intel Foundry, management indicated that key customers share their five-year production roadmaps with TSMC, implying TSMC will remain the primary source for customers' top products. We expect TSMC could increase leading-edge wafer pricing by another 5%–10% in 2027, given the value it provides in leading-edge foundry services. On gross margin guidance, the 3Q26 midpoint is 66% vs. MSe 67.5%, reflecting dilution from 2nm. While this falls below some unrealistic 70% Street forecasts, we reiterate our view to buy on any weakness post-Street margin forecast miss. TSMC consistently reminds investors about dilution from 2nm and overseas fabs. We do not believe the margin "miss" reflects foundry competition.

Raise PT to NT\$2,988; stay OW: We raise our 2026–2028 revenue assumptions to reflect stronger AI semiconductor revenue growth. TSMC's high-quality earnings profile should continue to attract fund flows in a volatile market environment. Upcoming 2Q26 cloud capex updates from CSPs are likely key catalysts.

MORGAN STANLEY TAIWAN LIMITED+

Charlie Chan

Equity Analyst

Charlie.Chan@morganstanley.com

+886 2 2730-1725

Daniel Yen, CFA

Equity Analyst

Daniel.Yen@morganstanley.com

+886 2 2730-2863

MORGAN STANLEY ASIA LIMITED+

Daisy Dai, CFA

Equity Analyst

Daisy.Dai@morganstanley.com

+852 2848-7310

MORGAN STANLEY TAIWAN LIMITED+

Tiffany Yeh

Equity Analyst

Tiffany.Yeh@morganstanley.com

+886 2 7712-3032

Lucas Wang

Research Associate

Lucas.Wang@morganstanley.com

+886 2 2730-2875



TSMC (2330.TW, 2330 TT)

Greater China Technology Semiconductors | Taiwan

Stock Rating	Overweight		
Industry View	Attractive		
Price target	NT\$2,988.00		
Up/downside to price target (%)	21		
Shr price, close (Jul 16, 2026)	NT\$2,470.00		
Mkt cap, curr (mn)	NT\$64,042,111		
Avg daily trading value (mn)	NT\$65,119		

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT\$)**	66.25	111.08	144.93	183.55
Prior EPS (NT\$)**	-	107.56	143.01	177.30
EPS (NT\$)§	64.56	100.37	130.74	166.68
ModelWare net inc (NT \$ bn)	1,718	2,874	3,758	4,759
P/E	23.4	22.3	17.0	13.5
Div yld (%)	1.4	1.1	1.4	1.7

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework

** = Based on consensus methodology

§ = Consensus data is provided by Refinitiv Estimates

e = Morgan Stanley Research estimates

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

2Q26 earnings and 3Q26 guidance review

TSMC hosted its 2Q26 earnings call on July 16. The company's 2Q26 results and 3Q26 guidance fall in line with our [Catalyst preview](#) Scenarios 1 and 2, where we assume the stock moves up 1%-5% over the coming trading days. See [Exhibit 1](#) and [Exhibit 2](#) for our review of the quarterly results and guidance.

2Q26 gross margin of 67.7% is roughly in line with MSe 67.4% and above the company's initial guidance range of 65.5%-67.5%. TSMC's 3Q26 revenue growth guidance of 12% Q/Q at the mid-point is also in line with our bullish forecast of 10-15% Q/Q. The flattish 3Q26 gross margin figure is in line with our preview, but lower than some bullish buy-side expectations of 70%. We therefore recommend investors buy on any weakness should the share price decline following the margin miss.

We now assume 2026-2028 three-year capex of US\$213bn vs. US\$206bn previously.

Exhibit 1: TSMC 2Q26 Earnings review

(NT\$ mn)	2Q26				
	Actual	Q/Q	Y/Y	MS Est.	Consensus
Financials					
Revenue	1,270,380	12.0%	36.0%	1,266,623	1,267,975
Opex	(93,708)	1.5%	11.6%	(96,667)	(109,471)
EPS (NT\$)	27.25	23.4%	77.4%	25.08	24.33
Ratios					
GM (%)	67.7%	147bps	910bps	67.4%	67.3%
Opex (%)	7.4%	-76bps	-161bps	7.6%	8.6%
OPM (%)	60.3%	224bps	1,072bps	59.8%	58.7%

Source: Company Data, Visible Alpha, Morgan Stanley Research estimates

Exhibit 2: TSMC 3Q26 Guidance review

(NT\$ mn)	3Q26				
	Guidance	MS Est.	Q/Q	Y/Y	Consensus
Financials					
Revenue	US\$44.6bn to US\$45.8bn, +12% Q/Q increase at mid-point	1,435,872	13.0%	45.0%	1,385,740
Opex		(104,250)	11.2%	18.7%	(116,721)
EPS (NT\$)		29.21	7.2%	67.5%	27.21
Ratios					
GM (%)	65.0% - 67.0%	67.5%	-20bps	807bps	66.4%
Opex (%)		7.3%	-12bps	-161bps	8.4%
OPM (%)	56.0% - 58.0%	60.3%	-8bps	969bps	58.0%

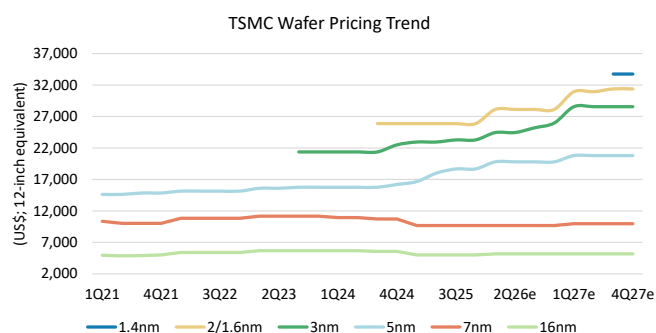
Source: Company Data, Visible Alpha, Morgan Stanley Research estimates

AI semi demand remains strong and requires additional capex

TSMC's 2026–2028 foundry floorplan broadly aligns with our bottom-up demand analysis

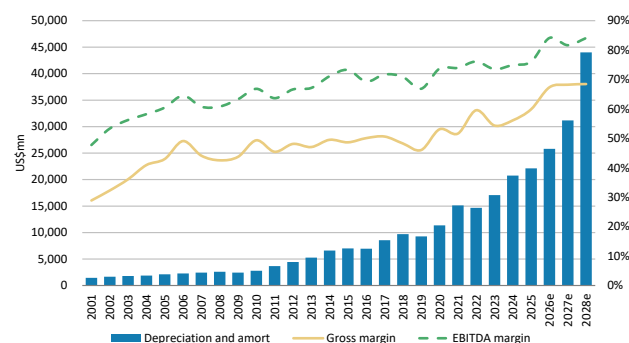
Our industry checks suggest ASML allocates capacity based largely on historical demand and long-term partnerships, indicating that TSMC remains the largest customer for EUV shipments. If so, TSMC should retain strong pricing power and could increase leading-edge wafer prices by 5%–10% in 2027, reflecting both the value it delivers to customers and the potential for margin expansion.

Exhibit 3: TSMC – Wafer pricing trend



Source: Company data, Morgan Stanley Research (e) estimates

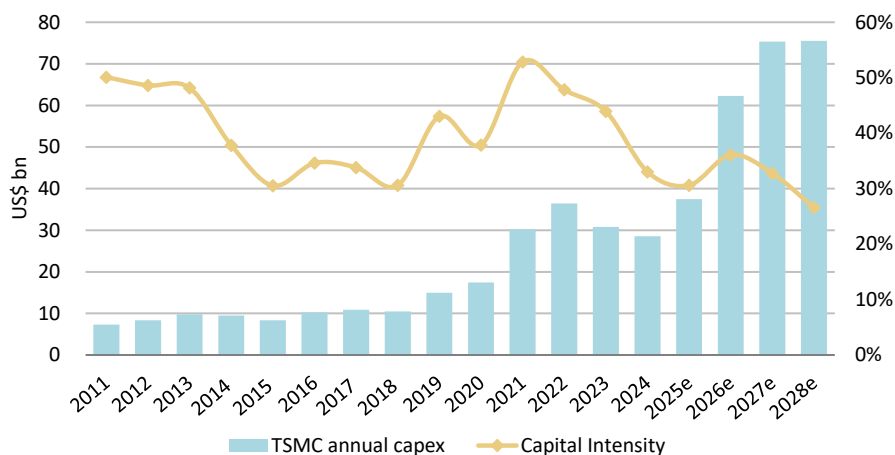
Exhibit 4: TSMC's GM and depreciation trends – 60% should be the floor



Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 5:

TSMC's capex intensity should drop significantly as a result of robust revenue growth



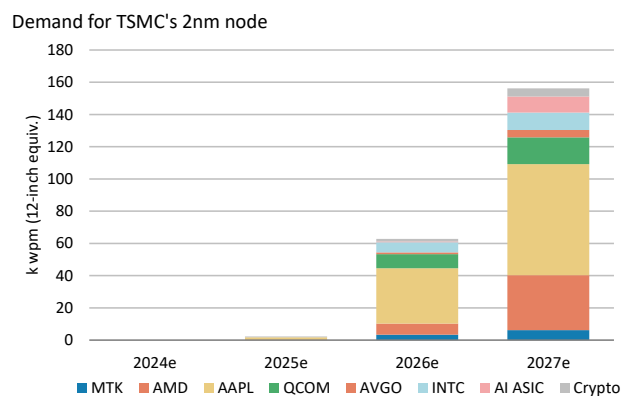
Source: Company data, Morgan Stanley Research estimates

TSMC revealed strong capacity growth in 2nm and 3nm with 5nm set to decline

Consistent with our demand forecast, TSMC indicated very strong capacity requirements for both 2nm and 3nm starting in 2026E, implying a 70% capacity CAGR for N2 from 2026E to 2028E. TSMC also suggested that 5nm capacity will begin to decline from 2027, in line with our forecast that 5nm demand will fall as Blackwell transitions to Rubin on 3nm.

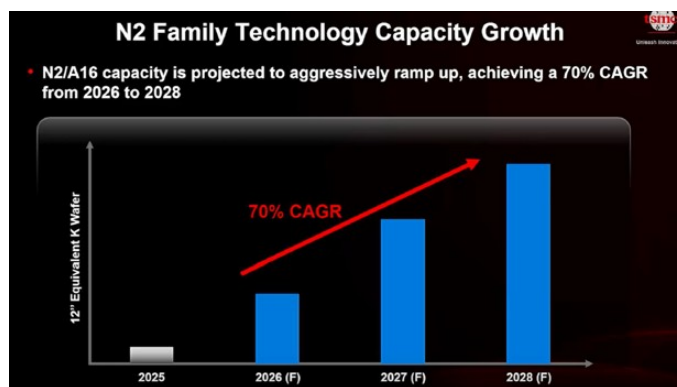
Separately, advanced packaging capacity expansion should remain a key focus for TSMC in 2027. This includes, but is not limited to, CoWoS and SoIC, for which we now expect capacity to reach 200kwpm and 70kwpm, respectively, by 2027.

Exhibit 6: TSMC – Our estimate of 2nm customer demand breakdown



Source: Morgan Stanley Research estimates

Exhibit 7: TSMC suggests N2 capacity can be >70% CAGR from 2026-2028e



Source: TSMC

Since nodes below 3nm should also continue to see strong demand, our checks suggest TSMC's 2nm/1.6nm capacity could approach 100kwpm by end-2026, increase to 150–170kwpm in 2027, and exceed 200kwpm in 2028. We expect 1.4nm to begin pulling in tools in 2027, with the bulk of capacity ramping in 2028. Looking further ahead, we expect TSMC to introduce initial 1.0nm capacity in Tainan in 2029.

Exhibit 8: TSMC fab roadmap

Node	Location	Fab	Capacity				
			2025	2026	2027	2028	2029
N3	Tainan	F18 - P4/P5/P6/P8/P9	110k	160-170k			
	Arizona	F21 - P2		20k			
	Japan	F23 - P2				15k	
Total			110k	180-190k	190k	200-205k	
N2/A16	HsinChu	F20 - P1/P2	20k	30k			
	Tainan	F22 - P7/P8/P9			40-50k	60k	
	KaoHsiung	F22 - P1/P2/P3/P4/P5/P6	25k	60k	80-90k	100k	
	Arizona	F21 - P3/P4/P5/P6				20k	
Total			45k	90-100k	150-170k	210k	
A14	HsinChu	F20 - P3/P4			10-20k	30-40k	40k
A14/A13/A12	Taichung	F25 - P1/P2/P3/P4			10-20k	30-40k	80k
Total					20-40k	60-80k	120k
A10	Tainan	F26 - P1/P2/P3/P4					5k
N40/28/22/12	Japan	F23 - P1	25k				
N28/16	Germany	F24			5k	40k	

Source: Morgan Stanley Research estimates

Exhibit 9: TSMC fab details

Front-end Plant	Location	Focused Technology
F18 - P4/P5/P6/P8/P9	Tainan	N3
F20 - P1/P2/P3/P4	HsinChu	N2/A14
F21 - P2	Arizona	N3
F22 - P1/P2/P3/P5/P6	KaoHsiung	N2
F22 - P7/P8/P9	Tainan	N2
F22 - P4	KaoHsiung	A16
F23 - P2	Kumamoto	N3
F25 - P1/P2/P3/P4	Taichung	A14
F26 - P1/P2/P3/P4	Tainan	A10

Source: Morgan Stanley Research estimates

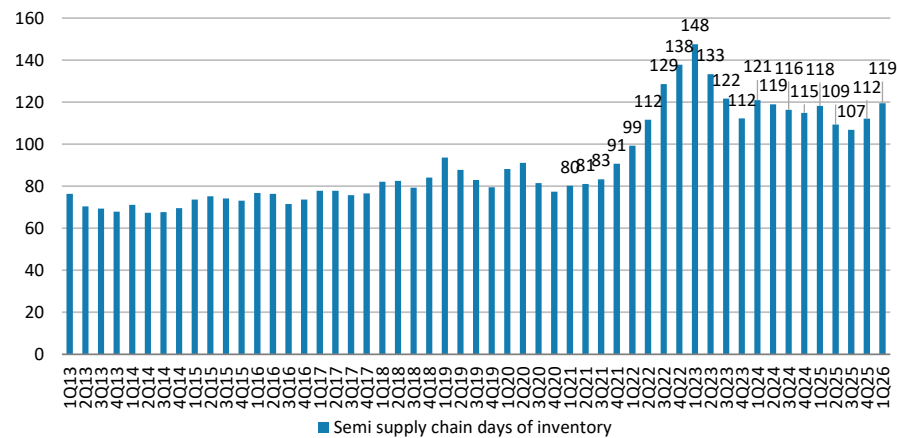
Non-AI demand – China's smartphone chip inventory correction begins, but the impact on TSMC should be limited

In 1Q26, we saw semiconductor industry inventory days increase amid strong AI and peripheral demand. Some consumer and non-AI customers also refrained from cutting orders given tight capacity across the supply chain and the prospect of foundry price increases in 2H26 and 2027. However, we believe it is now time for these non-AI customers, particularly in smartphones and PCs, to address rising BOM costs either by reducing orders or passing through higher costs. Even so, we expect the impact on TSMC to remain limited, as its consumer customer base is concentrated in the premium segment, where passing through higher costs is generally easier. In addition, high-end PCs and smartphones primarily consume leading-edge capacity at TSMC, which AI demand can readily absorb once any wafer capacity becomes available.

Apple provides an example of successful price increases to end customers (see [Apple, Inc.: As Expected, Apple Is Raising Product Pricing \(25 Jun 2026\)](#) by Erik Woodring) while maintaining the same wafer demand at TSMC based on our checks.

In contrast, Chinese smartphone shipments are tracking at down 20% Y/Y, compared with our previous assumption of down 15% Y/Y (see our [MediaTek report](#)). Even so, we expect AI demand to quickly backfill any leading-edge wafer capacity, particularly at 5nm and below.

Exhibit 10: Semi supply chain days of inventory piled up in 1Q26



Source: FactSet, Morgan Stanley Research

Earnings estimate revisions

We raise our earnings forecasts 3% for 2026, 1% for 2027 and 4% for 2028: We raise our 2026 revenue forecast and modestly increase our margin assumptions to reflect stronger-than-expected 2Q26 performance, driven by robust AI demand, including sustained Blackwell shipments. We also raise our 2027–2028 revenue assumptions by 1%–3%, reflecting the increase in 2026 capex guidance from US\$54bn to US\$62bn at the midpoint.

Exhibit 11: TSMC: Earnings revision

(NT\$ mn)	New 2026e	Old 2026e	Diff. %	New 2027e	Old 2027e	Diff. %	New 2028e	Old 2028e	Diff. %
Net sales	5,438,831	5,360,940	1%	7,187,423	7,132,285	1%	9,037,662	8,787,896	3%
Gross profit	3,663,403	3,598,737	2%	4,912,038	4,851,218	1%	6,219,699	6,023,802	3%
Operating profit	3,249,517	3,196,368	2%	4,342,910	4,283,717	1%	5,548,069	5,356,165	4%
Pretax income	3,425,902	3,302,788	4%	4,446,359	4,387,166	1%	5,647,518	5,455,614	4%
Net income	2,880,237	2,789,152	3%	3,758,166	3,708,187	1%	4,759,400	4,597,307	4%
Diluted EPS	111.08	107.56	3%	144.93	143.01	1%	183.55	177.30	4%
Margins									
Gross margin	67.4%	67.1%		68.3%	68.0%		68.8%	68.5%	
Operating margin	59.7%	59.6%		60.4%	60.1%		61.4%	60.9%	
Pretax margin	63.0%	61.6%		61.9%	61.5%		62.5%	62.1%	
Net margin	53.0%	52.0%		52.3%	52.0%		52.7%	52.3%	

Source: Morgan Stanley Research estimates

Price target raised to NT\$2,988

Our revised price target of NT\$2,988 is driven primarily by earnings estimate revisions. It implies a 20x P/E multiple on our new 2027E EPS forecast. TSMC currently trades at 17x our 2027E EPS estimate, close to its average NTM P/E of 16.5x since 2018. We find the stock attractive at current levels.

We continue to derive our 12-month price target (base-case scenario) using a residual income valuation framework. Our key assumptions remain unchanged, including a cost of equity of 9.2% (beta of 1.2, equity risk premium of 6.0%, and risk-free rate of 2.0%), an intermediate growth rate of 10.5%, and a terminal growth rate of 4%.

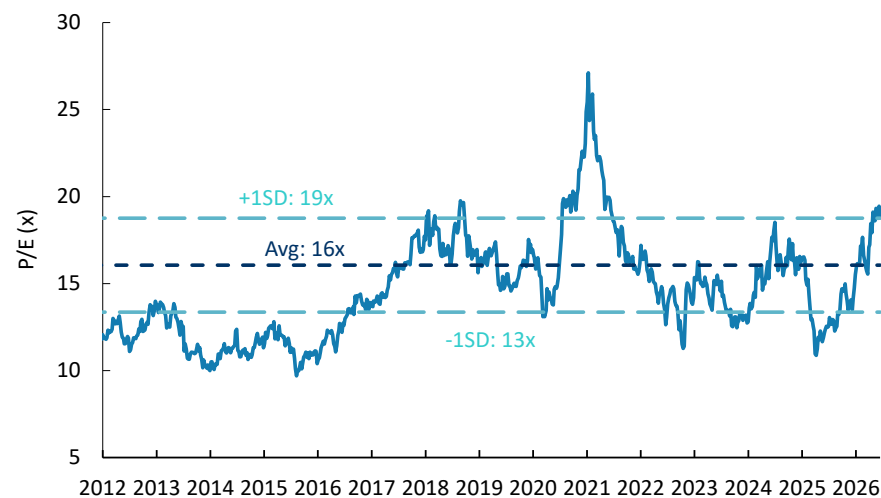
We also raise our bull-case and bear-case valuations to NT\$3,600 (from NT\$3,480) and NT\$1,645 (from NT\$1,590), respectively.

Exhibit 12: TSMC: Residual income model

	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e
Total Equity	7,799,296	10,805,424	14,605,326	17,971,174	21,690,436	25,800,221	30,341,533	35,359,683	40,904,738	47,032,024	53,802,675	61,284,245
Net Profit	2,880,237	3,758,166	4,759,400	5,259,137	5,811,347	6,421,538	7,095,800	7,840,859	8,664,149	9,573,885	10,579,143	11,689,953
ROAE	43.4%	40.4%	37.5%	32.3%	29.3%	27.0%	25.3%	23.9%	22.7%	21.8%	21.0%	20.3%
Residual Income	1,869,896	2,433,391	3,053,585	3,372,066	3,613,041	3,870,306	4,148,200	4,450,566	4,781,103	5,143,557	5,541,849	5,980,166
Spread	34.2%	31.2%	28.3%	23.1%	20.1%	17.8%	16.1%	14.7%	13.5%	12.6%	11.8%	11.1%
Ending Equity Capital	7,799,296											
PV of Forecast Period	24,254,430											
PV of Continuing Value	45,424,925											
Equity Value	77,478,651											
No. of Shares	25,930											
Projected Price (NT\$)	2,988											

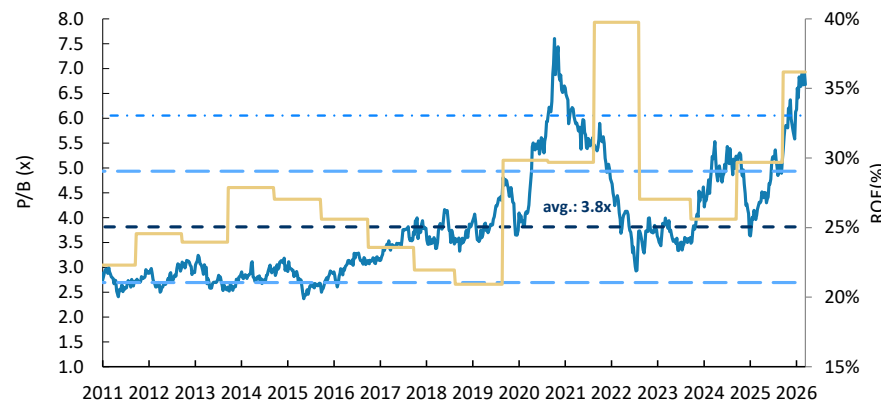
Source: Company data, Morgan Stanley Research (e) estimates

Exhibit 13: TSMC's one-year forward P/E: Valuation does not appear stretched as a key AI semi enabler



Source: Company data, FactSet, Morgan Stanley Research

Exhibit 14: TSMC – P/B vs. ROE



Source: Company data, Refinitiv, Morgan Stanley Research

Exhibit 15: TSMC: Quarterly financials

(NT\$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	2023	2024	2025	2026e	2027e	2028e
Total Revenues	839,254	933,792	989,918	1,046,090	1,134,103	1,270,380	1,470,880	1,563,467	2,161,736	2,894,308	3,809,054	5,438,831	7,187,423	9,037,662
Sequential Change	-3.4%	11.3%	6.0%	5.7%	8.4%	12.0%	15.8%	6.3%						
Change vs Year Ago	41.6%	38.6%	30.3%	20.5%	35.1%	36.0%	48.6%	49.5%	-4.5%	33.9%	31.6%	42.8%	32.2%	25.7%
Cost of Sales	(345,859)	(386,423)	(401,375)	(394,104)	(382,808)	(410,070)	(477,897)	(502,894)	(986,625)	(1,269,954)	(1,527,760)	(1,773,669)	(2,275,385)	(2,817,963)
Percent of Revenues	41.2%	41.4%	40.5%	37.7%	33.8%	32.3%	32.5%	32.2%	45.6%	43.9%	40.1%	32.6%	31.7%	31.2%
Gross Profit	493,395	547,369	588,543	651,987	751,295	860,310	992,983	1,060,573	1,175,111	1,624,354	2,281,294	3,663,403	4,912,038	6,219,699
Gross Margin	58.8%	58.6%	59.5%	62.3%	66.2%	67.7%	67.5%	67.8%	54.4%	56.1%	59.9%	67.4%	68.3%	68.8%
Total Opex	(86,314)	(83,946)	(87,858)	(87,084)	(92,329)	(93,708)	(108,280)	(113,240)	(253,645)	(302,301)	(345,202)	(413,885)	(569,129)	(671,630)
Percent of Revenues	10.3%	9.0%	8.9%	8.3%	8.1%	7.4%	7.4%	7.2%	11.7%	10.4%	9.1%	7.6%	7.9%	7.4%
R&D	(56,547)	(61,280)	(63,742)	(64,858)	(67,757)	(73,146)	(77,146)	(80,146)	(182,370)	(204,182)	(246,427)	(298,195)	(356,300)	(400,500)
Percent of Revenues	6.7%	6.6%	6.4%	6.2%	6.0%	5.8%	5.2%	5.1%	8.4%	7.1%	6.5%	5.5%	5.0%	4.4%
SG&A	(29,767)	(22,666)	(24,116)	(22,227)	(24,573)	(26,890)	(31,134)	(33,094)	(71,275)	(98,119)	(98,775)	(115,691)	(212,829)	(271,130)
Sequential Change	1.1%	-23.9%	6.4%	-7.8%	10.6%	9.4%	15.8%	6.3%						
Change vs Year Ago	53.6%	-3.6%	-6.5%	-24.5%	-17.4%	18.6%	29.1%	48.9%						
Operating Income	407,081	463,424	500,685	564,902	658,966	766,602	884,703	947,333	921,466	1,322,053	1,936,092	3,249,517	4,342,910	5,548,069
Percent of Revenues	48.5%	49.6%	50.6%	54.0%	58.1%	60.3%	60.1%	60.6%	42.6%	45.7%	50.8%	59.7%	60.4%	61.4%
Non-operating Income(Loss)	23,815	29,612	24,684	27,461	28,834	95,827	25,862	25,862	57,706	83,786	105,571	176,385	103,449	99,449
Profit Before Taxes	430,895	493,035	525,369	592,363	687,800	862,429	910,565	973,195	979,171	1,405,839	2,041,663	3,425,902	4,446,359	5,647,518
Percent of Revenues	51.3%	52.8%	53.1%	56.6%	60.6%	67.9%	61.9%	62.2%	45.3%	48.6%	53.6%	63.0%	61.9%	62.5%
Taxes	(70,163)	(95,542)	(73,614)	(86,948)	(114,998)	(155,649)	(136,585)	(145,979)	(141,404)	(233,407)	(326,266)	(553,211)	(688,192)	(888,118)
Tax Rate	16.3%	19.4%	14.0%	14.7%	16.7%	18.0%	15.0%	15.0%	14.4%	16.6%	16.0%	16.1%	15.5%	15.7%
Net Income, Cont Ops	360,733	397,493	451,755	505,415	572,801	706,780	773,980	827,216	837,768	1,172,432	1,715,397	2,872,691	3,758,166	4,759,400
Minority Interest	831	780	546	329	(322)	(219)	0	0	730	836	2,486	(541)	0	0
Reported Income (TW GAAP)	361,564	398,273	452,301	505,744	572,480	706,561	773,980	827,216	838,498	1,173,268	1,717,883	2,880,237	3,758,166	4,759,400
Percent of Revenues	43.1%	42.7%	45.7%	48.3%	50.5%	55.6%	52.6%	52.9%	38.8%	40.5%	45.1%	53.0%	52.3%	52.7%
Change vs Year Ago	60.3%	60.7%	39.1%	35.0%	58.3%	77.4%	71.1%	63.6%	-17.5%	39.9%	46.4%	67.7%	30.5%	26.6%
Reported EPS (NT\$, TW GAAP)	13.94	15.36	17.44	19.50	22.08	27.25	29.85	31.90	32.34	45.25	66.25	111.08	144.93	183.55
Basic shares (mn units)	25,927	25,928	25,929	25,928	25,930	25,930	25,930	25,930	25,929	25,928	25,928	25,930	25,930	25,930

Source: Company data, Morgan Stanley Research (e) estimates

Risk Reward – TSMC (2330.TW)

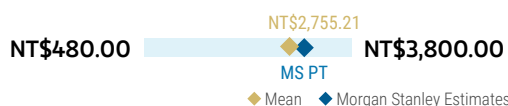
2026 revenue and capex guidance raised; OW

PRICE TARGET NT\$2,988.00

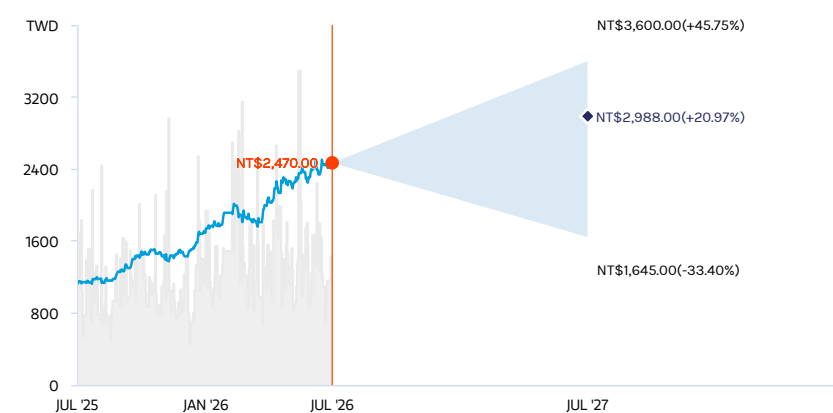
Base case, residual income model. Key assumptions: a cost of equity of 9.2% (beta of 1.2, risk-free rate of 2.0% and risk premium of 6.0%), an intermediate growth rate of 10.5%, and a terminal growth rate of 4.0%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



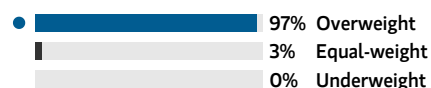
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Reflecting robust AI capex guidance from Meta and Microsoft, TSMC remains our Top Pick.
- Our estimates reflect TSMC's strong profitability and another year of close to 40% Y/Y revenue growth seen for 2026.
- Part of that can be contributed by China's AI GPU market through Nvidia.
- Reflecting increased bargaining power, long-term margin expansion, and sustainable AI semi demand, we expect the stock to re-rate to our implied 2027e target P/E of 20x.

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*
Pricing Power: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$3,600.00

24x 2027e EPS

TSMC dominates foundry services: 1) Breakthrough in EUV tech and materials accelerates node migration to 3nm and beyond. 2) Intel outsources its server CPU production to TSMC sooner than expected. 3) Intel or Samsung exits leading-edge foundry business. 4) New tech megatrends such as 6G or AI drive global semi revenue growth and increase demand for leading edge nodes. 5) Cost per transistor decreases further along with node migration.

BASE CASE

NT\$2,988.00

20x 2027e EPS

Leading position in leading-edge logic foundry services with some gross margin expansion: 1) Global semi revenue CAGR is sustained in the high single digits in 2025-28. 2) TSMC continues to lead in market share in 16nm, 7nm, 5nm, 3nm, and 2nm. 3) It continues to observe robust AI-related demand. 4) Cost per transistor does not decrease meaningfully in 3nm and 2nm, while capex per k is more demanding than for 7nm and 5nm.

BEAR CASE

NT\$1,645.00

11x 2027e EPS

TSMC's dominance in foundry business erodes: 1) Global ex-memory semi revenue growth is weaker than expected in 2025-28. 2) Declining trailing-edge market share causes revenue shortfall. 3) Intel's contribution is weaker than expected. Intel and/or Samsung successfully develop leading-edge process technology. 4) Intel's foundry business execution proves successful and weighs on TSMC with order competition. 5) Demand for 2nm is less than expected given high transistor cost for foundry customers.

Risk Reward – TSMC (2330.TW)

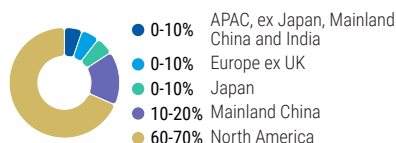
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Revenue from 5nm Geometry (NT\$, mn)	1,179,907	1,351,242	1,118,627	1,111,915
Revenue from 10/7nm Geometry (NT\$, mn)	459,258	516,628	525,849	523,809
Revenue from 16/20nm Geometry (NT\$, mn)	220,027	292,481	318,988	336,404

INVESTMENT DRIVERS

- Moore's Law migration (chip scaling) and alternative technologies
- Capex intensity - cost of building leading-edge foundry capacity
- Bargaining power with key equipment vendors
- Transistor cost for customers
- ROI for leading-edge foundry

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

1/5
MOST 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- TSMC charges large customers more to keep its gross margin above 56% in the long term.
- AI semi demand grows more significantly than expected, while TSMC maintains high market share in leading-edge foundry business.
- Outsourcing from Intel CPU increases in 2025-27.

RISKS TO DOWNSIDE

- Inventory correction occurs in 2026.
- Demand for leading edge technologies weakens.
- Costs of overseas fabs grow significantly.

OWNERSHIP POSITIONING

Inst. Owners, % Active 75.9%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT\$, mn) 5,026,000 5,438,831 5,494,192
5,229,946

EBITDA (NT\$, mn) 3,295,000 4,055,625 4,157,645
3,838,766

Net income (NT\$, mn) 2,369,115 2,596,362 2,873,909

EPS (NT\$) 89.80 100.37 111.08

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Financial summary

Income Statement, 2024-2028e, Year End Dec

(NT\$ mn)	2024	2025	2026e	2027e	2028e
Revenue	2,894,308	3,809,054	5,438,831	7,187,423	9,037,662
YoY Growth	33.9%	31.6%	42.8%	32.2%	25.7%
Gross Profit b/f dep	2,247,086	2,924,669	4,413,759	5,829,809	7,516,251
Less: COGS	(1,269,954)	(1,527,760)	(1,773,669)	(2,275,385)	(2,817,963)
Variable costs	(647,221)	(884,385)	(1,025,072)	(1,357,614)	(1,521,411)
Depreciation & amort	(622,733)	(643,376)	(750,356)	(917,771)	(1,296,552)
Gross profit	1,624,354	2,281,294	3,663,403	4,912,038	6,219,699
% margin	56.1%	59.9%	67.4%	68.3%	68.8%
YoY Growth	38.2%	40.4%	60.6%	34.1%	26.6%
Operating Expenses:	(302,301)	(345,202)	(413,885)	(569,129)	(671,630)
R&D	(204,182)	(246,427)	(298,195)	(356,300)	(400,500)
Sales and Marketing	(13,144)	(16,918)	(21,175)	(35,937)	(45,188)
General and Admin	(84,975)	(81,857)	(94,516)	(176,892)	(225,942)
Operating Profit	1,322,053	1,936,092	3,249,517	4,342,910	5,548,069
% margin	45.7%	50.8%	59.7%	60.4%	61.4%
YoY Growth	43.5%	46.4%	67.8%	33.6%	27.8%
Non-Operating Income	83,786	105,571	176,385	103,449	99,449
Pretax Profit	1,405,839	2,041,663	3,425,902	4,446,359	5,647,518
% margin	48.6%	53.6%	63.0%	61.9%	62.5%
Tax	(233,407)	(326,266)	(553,211)	(688,192)	(888,118)
Reported net Income	1,173,268	1,717,883	2,880,237	3,758,166	4,759,400
Reported EPS (NT\$)	45.25	66.25	111.08	144.93	183.55
EPS for consensus (NT\$)	45.25	66.25	111.08	144.93	183.55

Key Ratios, 2024-2028e

	2024	2025	2026e	2027e	2028e
Return (%)					
ROAA	19%	23%	31%	31%	31%
ROAE	30%	35%	44%	41%	38%
OP. ATO	0.7x	0.7x	0.8x	1.0x	0.9x
Gearing (x)					
Net Debt/ Equity	-0.33x	-0.38x	-0.40x	-0.42x	-0.49x
Current Ratio	2.4x	2.6x	3.0x	3.9x	5.4x
Quick Ratio	1.9x	2.1x	2.5x	3.3x	4.8x
Operating Cycle					
AR/NR Turnover (days)	29	26	24	25	25
Inventory Turnover (days)	76	68	69	70	69
AP Turnover (days)	19	19	21	21	21
Cash Conversion (days)	87	75	72	74	73

e = Morgan Stanley Research Estimates
Source: Company Data, Morgan Stanley Research

Balance Sheet, 2024-2028e, Year End Dec

(NT\$ mn)	2024	2025	2026e	2027e	2028e
Cash & Equivalent	2,127,627	2,767,856	3,810,923	5,252,578	7,883,171
Marketable Security	294,393	300,738	347,964	347,964	347,964
A/R & N/R	272,088	281,791	443,238	571,072	691,480
Inventories	287,869	288,109	392,744	489,357	593,438
Other Current Ass.	106,376	178,635	207,452	207,452	207,452
Total current assets	3,088,352	3,817,131	5,202,321	6,868,424	9,723,505
Long-term Investment	149,040	172,370	166,567	166,567	166,567
Total fixed assets	3,234,980	3,691,841	4,905,905	6,275,690	7,252,573
Total other assets	219,565	251,682	274,192	274,192	274,192
Total Assets	6,691,938	7,933,024	10,548,984	13,584,872	17,416,837
A/P & N/P	74,227	84,330	120,983	150,744	182,806
Accrued Expenses	0	0	0	0	0
Other Payable	531,377	630,217	754,314	754,314	754,314
Curr. of L-T Debt	59,858	136,926	156,242	156,242	156,242
Other Curr. Liab.	599,064	606,547	703,842	703,842	703,842
Total Current Liab.	1,264,525	1,458,019	1,735,380	1,765,141	1,797,203
L-T Liabilities	926,605	856,228	860,026	860,026	860,026
Total Other L-T Liab	177,233	157,982	154,281	154,281	154,281
Total Liabilities	2,368,362	2,472,229	2,749,687	2,779,449	2,811,510
Common Stocks	259,327	259,325	259,324	259,324	259,324
Preferred Stocks	0	0	0	0	0
Capital Reserve	73,261	73,446	73,666	73,666	73,666
Retained earnings	3,955,957	5,086,825	7,424,878	10,431,005	14,230,908
Treasury Stock	0	0	0	0	0
Minority Equity	35,031	41,199	41,429	41,429	41,429
Total Equity	4,323,576	5,460,795	7,799,296	10,805,424	14,605,326
Total Liab. & Equity	6,691,938	7,933,024	10,548,984	13,584,872	17,416,837

Cash Flow Statement, 2024-2028e, Year End Dec

(NT\$ mn)	2024	2025	2026e	2027e	2028e
Net Income Current	1,173,268	1,717,883	2,880,237	3,758,166	4,759,400
Depreciation	662,797	688,096	804,349	966,075	1,364,792
Net Investment Losses (Gains)	8,205	8,083	5,115	0	0
Decrease (Increase) in Working Capital	20,896	(17,144)	(63,063)	(194,686)	(192,427)
Others	(38,988)	(121,942)	(121,402)	0	0
Cash Flow-Operating	1,826,177	2,274,976	3,505,236	4,529,555	5,931,765
(Purchase) of FA	(956,007)	(1,272,411)	(1,940,886)	(2,335,860)	(2,341,675)
Sale of Fix Asset	895	797	1,813	0	0
(Purchase)L-T Inv.	(3,739)	0	0	0	0
Others	94,008	127,220	(7,904)	0	0
Cash Flow-Inv.	(864,843)	(1,144,393)	(1,946,977)	(2,335,860)	(2,341,675)
Inc(Dec)-S-T Debt	0	0	0	0	0
Inc(Dec) L-T Debt	28,601	7,948	1,630	0	0
Dividend Paid	(363,055)	(466,779)	(570,513)	(752,039)	(959,498)
Dir.&Emp.Bonus	0	0	0	0	0
Others	(11,847)	18,486	8,122	0	0
Cash Flow-Financing	(346,301)	(440,345)	(560,761)	(752,039)	(959,498)
Change in Cash	662,199	640,229	1,043,066	1,441,656	2,630,592
Net cash/(debt), b/f	1,465,428	2,127,627	2,767,856	3,810,923	5,252,578
Net cash/(debt), c/f	2,127,627	2,767,856	3,810,923	5,252,578	7,883,171

Disclaimer: This report references U.S. Executive Order 14032 and/or entities or securities that are designated thereunder. U.S. persons may be prohibited from buying certain securities of entities named in this report. Readers are solely responsible for ensuring that their investment activities are carried out in compliance with applicable laws.

Morgan Stanley is acting as exclusive financial advisor to AP Grange Holdings, LLC ("AP Grange") in relation to a definitive agreement with Intel Corporation ("Intel"), for Intel to repurchase the 49% equity interest in the joint venture related to Intel's Fab 34 in Ireland not held by Intel, as announced on April 1, 2026. The transaction is subject to customary closing conditions. AP Grange has agreed to pay fees to Morgan Stanley for its financial advisory services. Please refer to the notes at the end of this report.

Morgan Stanley is acting as financial advisor to Grail, Inc. ("Grail") in connection with the proposed strategic collaboration with Samsung C&T and Samsung Electronics, to bring Grail's Galleri multi-cancer early detection test to key Asian markets, as announced on October 16, 2025. The proposed transaction is subject to final execution of definitive collaboration agreements between the parties, as well as customary closing conditions and regulatory approvals. Please refer to the notes at the end of this report.

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Charlie Chan, CFA; Daniel Yen, CFA.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of June 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: ACM Research Inc, Advanced Micro-Fabrication Equipment Inc, Advanced Wireless Semiconductor Co, Alchip Technologies Ltd, AllRing Tech Co., AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMPT Ltd, Cambricon Technology Corporation, Dosilicon Co Ltd, FOCI Fiber Optic Communications Inc, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Hangzhou Silan Microelectronics Co. Ltd., King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, Montage Technology Co Ltd, Parade Technologies Ltd, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, SG Micro Corp., Shanghai Fudan Microelectronics, Silergy Corp., Silicon Motion, **TSMC**, UMC, Unigroup Guoxin Microelectronics Co Ltd, Vanguard International Semiconductor, WIN Semiconductors Corp, Winbond Electronics Corp, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Alchip Technologies Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from ASMPT Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Micro-Fabrication Equipment Inc, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Motion, **TSMC**, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from ASE Technology Holding Co. Ltd., King Yuan Electronics Co Ltd, MediaTek, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Silicon Motion, SMIC, **TSMC**, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Micro-Fabrication Equipment Inc, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Motion, **TSMC**, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: ASE Technology Holding Co. Ltd., King Yuan Electronics Co Ltd, MediaTek, Montage Technology Co Ltd, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Silicon Motion, SMIC, **TSMC**, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not

linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of June 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

TSMC (2330.TW) - As of 07/16/26 GMT in TWD
Industry : Greater China Technology Semiconductors



Stock Rating History: 7/1/21 : E/I; 10/12/21 : E/C; 2/1/22 : O/C; 10/4/22 : O/A; 2/22/23 : O/I; 7/7/23 : O/A; 7/21/24 : O/I; 7/14/25 : O/A

Price Target History: 6/18/21 : 580; 1/3/22 : 618; 1/13/22 : 648; 2/7/22 : 780; 9/23/22 : 720; 12/14/22 : 700; 4/20/23 : 680; 6/12/23 : 710; 7/7/23 : 718; 12/4/23 : 688; 1/18/24 : 698; 2/14/24 : 758; 3/8/24 : 850; 3/22/24 : 860; 5/8/24 : 928; 6/6/24 : 980; 6/18/24 : 1080; 7/8/24 : 1180; 7/18/24 : 1220; 9/24/24 : 1280; 10/17/24 : 1330; 12/16/24 : 1388; 4/11/25 : 1288; 7/17/25 : 1388; 9/30/25 : 1588; 10/16/25 : 1688; 12/18/25 : 1888; 1/14/26 : 1988; 1/16/26 : 2088; 3/9/26 : 2288; 4/17/26 : 2588; 6/29/26 : 2888

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FCO118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital

Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of ASMPT Ltd, Hua Hong Semiconductor Ltd, Montage Technology Co Ltd listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/16/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$91.96
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb361.42
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$127.00
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$3,770.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$682.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,279.22
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT\$4,185.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$1,370.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$516.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$157.40
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$556.50
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$318.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb83.25
MediaTek (2454.TW)	O (11/28/2025)	NT\$3,700.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb890.00
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$439.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb702.76
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb109.85
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$1,980.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb107.32
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$473.00
SMIC (0981.HK)	O (10/21/2025)	HK\$75.20
TSMC (2330.TW)	O (02/07/2022)	NT\$2,470.00
UMC (2303.TW)	O (05/19/2026)	NT\$160.00
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$187.50
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$369.00

Daisy Dai, CFA

ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$177.50
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb65.50
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb106.21
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb36.22
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb320.40
Innoscence (2577.HK)	E (10/13/2025)	HK\$48.00
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb83.21
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$28.10
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb120.24
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb97.93
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb70.16
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb26.10
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb101.70

Daniel Yen, CFA

AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$909.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,515.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$13,905.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$101.50
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$175.50
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb120.61
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb514.61
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$136.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$278.60
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb211.13
Novatek (3034.TW)	U (02/04/2026)	NT\$471.50
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$156.50
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$631.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$76.50
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$712.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb54.83
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$172.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$113.50
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$196.00

Duan Liu

Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb136.50
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb463.59

Tiffany Yeh

AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,075.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$555.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$13.92
Hon Precision (7769.TW)	O (04/17/2026)	NT\$6,695.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$6,220.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$293.40
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$6,735.00

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.